

SRI SATHYA SAI INSTITUTE OF ACTUARIES

(SSSIA)

DETAILED SYLLABUS

AI Actuarial Courses in Actuarial Science

SSSIA Programs Aligned to International Education Qualification Framework

Certificate Level Equivalent – 40 Credits

Diploma Level Equivalent – 80 Credits

Graduation Level Equivalent – 120 Credits

All Programs Aligned to the Casualty Actuarial Society Associateship (ACAS) Requirements.



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CREDIT FRAMEWORK

Credits are assigned on the basis set out in the National Higher Education Qualifications Framework. One credit of a lecture or tutorial course means one hour of engagement per week, which over a semester of fifteen weeks is fifteen hours of instruction. One credit of practicum, laboratory work, project work or internship means two hours of engagement per week, which over the same fifteen weeks is thirty hours of engagement.

Accordingly, a four credit theory course carries sixty hours and a three credit practical course carries ninety hours. Total hours shown for each course in this document follow that rule without exception.

SCHEME OF INSTRUCTION

Code	Title of the Course	Type	Credits	Total Hours
PROBABILITY				
SSSIA-101	Probability	Theory	4	60
FINANCIAL MATHEMATICS				
SSSIA-102	Financial Mathematics	Theory	4	60
BUSINESS ECONOMICS				
SSSIA-103	Business Economics - Part I (Microeconomics)	Theory	3	45
SSSIA-104	Business Economics - Part II (Macroeconomics)	Theory	3	45
ACCOUNTING AND FINANCE				
SSSIA-105	Corporate Finance	Theory	3	45
SSSIA-106	Financial and Insurance Accounting	Theory	3	45
MODERN ACTUARIAL STATISTICS I				
SSSIA-107	Modern Actuarial Statistics I - Part I	Theory	4	60
SSSIA-108	Modern Actuarial Statistics I - Part II	Theory	4	60
MODERN ACTUARIAL STATISTICS II				
SSSIA-109	Modern Actuarial Statistics II - Part I	Theory	4	60
SSSIA-110	Modern Actuarial Statistics II - Part II	Theory	4	60
RATEMAKING				
SSSIA-111	Ratemaking - Part I	Theory	4	60
SSSIA-112	Ratemaking - Part II	Theory	4	60
RESERVING				
SSSIA-113	Reserving - Part I	Theory	4	60
SSSIA-114	Reserving - Part II	Theory	4	60
REGULATION AND FINANCIAL REPORTING				

SSSIA-115	Regulation and Financial Reporting - Part I	Theory	5	75
SSSIA-116	Regulation and Financial Reporting - Part II	Theory	5	75
PREDICTIVE ANALYTICS				
SSSIA-117	Predictive Analytics - Part I	Theory	6	90
SSSIA-118	Predictive Analytics - Part II	Theory	6	90
SSSIA-119	Predictive Analytics Project Practicum	Practical	4	120
AGENTIC AI APPLICATIONS				
SSSIA-120	Agentic AI for Probability	Practical	3	90
SSSIA-121	Agentic AI for Financial Mathematics	Practical	3	90
SSSIA-122	Agentic AI for Modern Actuarial Statistics I	Practical	3	90
SSSIA-123	Agentic AI for Modern Actuarial Statistics II	Practical	3	90
SSSIA-124	Agentic AI for Ratemaking	Practical	4	120
SSSIA-125	Agentic AI for Reserving	Practical	4	120
SSSIA-126	Agentic AI for Regulation and Reporting	Practical	4	120
EXCEL FOR ACTUARIAL PRACTICE				
SSSIA-127	Excel for Actuarial Practice - Part I	Practical	3	90
SSSIA-128	Excel for Actuarial Practice - Part II	Practical	3	90
R FOR ACTUARIAL WORK				
SSSIA-129	R for Actuarial Work - Part I	Practical	3	90
SSSIA-130	R for Actuarial Work - Part II	Practical	3	90
PYTHON FOR ACTUARIAL WORK				
SSSIA-131	Python for Actuarial Work - Part I	Practical	3	90
SSSIA-132	Python for Actuarial Work - Part II	Practical	3	90
TOTAL — 32 courses (Theory 74 credits, Practical 46 credits)			120	2490

DETAILED SYLLABI

SSSIA-101 Probability

SSSIA-101	Probability	CREDITS: 4 Total Hours: 60
Type of Course	Theory	15 hours per credit

Course Objectives

The student will understand and be able to:

- (a) basic probability concepts, combinatorics and discrete mathematics;
- (b) discrete and continuous univariate random variables, including the binomial, negative binomial, geometric, hypergeometric, Poisson, uniform, exponential, gamma, normal, lognormal and beta distributions, and their applications;
- (c) multivariate discrete random variables, the distribution of order statistics and linear combinations of independent random variables.

Course Syllabus

At the end of the course, the student will be able to:

Unit	Description	Hours
I	1. Define set functions, Venn diagrams, sample space and events. Define probability as a set function on a collection of events and state the basic axioms of probability. 2. Calculate probabilities using the addition and multiplication rules. 3. Define independence and calculate probabilities of independent events. 4. Calculate probabilities of mutually exclusive events. 5. Define and calculate conditional probabilities.	12
II	6. Calculate probabilities using combinatorics, such as combinations and permutations. 7. State Bayes' theorem and the law of total probability and use them to calculate conditional probabilities. 8. Explain and apply the concepts of random variables, probability density functions and cumulative distribution functions. 9. Explain and calculate expected value and higher moments, mode, median and percentile.	12
III	10. Explain and calculate variance, standard deviation and coefficient of variation. 11. Apply the concepts of deductibles, coinsurance, benefit limits and inflation to convert a given loss amount into the corresponding payment amount for an insurance company. 12. Calculate the expected value, variance and standard deviation of both the loss random variable and the corresponding payment random variable after policy adjustments. 13. Determine the sum of independent random variables for the Poisson and normal cases.	12
IV	14. Explain and perform calculations concerning joint probability functions and joint cumulative distribution functions. 15. Determine conditional and marginal probability functions. 16. Calculate moments for joint, conditional and marginal distributions. 17. Calculate variance and standard deviation for conditional and marginal probability distributions. 18. Calculate joint moments, such as covariance and the correlation coefficient.	14
V	19. Determine the distribution of order statistics from a set of independent random variables.	10

	20. Calculate probabilities for linear combinations of independent normal random variables. 21. Calculate moments for linear combinations of independent random variables. 22. Apply the central limit theorem to calculate probabilities for linear combinations of independent and identically distributed random variables.	
	Total	60

References

- Ross, S. M. (2019). A First Course in Probability, 10th ed. Pearson. ISBN 978-0134753119.
- Wackerly, D., Mendenhall III, W. and Scheaffer, R. (2008). Mathematical Statistics with Applications, 7th ed. Thomson Brooks/Cole. ISBN 978-0495110811.
- Hassett, M., Stewart, D. and Milovanovic, J. (2021). Probability for Risk Management, 3rd ed. ACTEX. ISBN 978-1-64756-322-6.
- Asimow, L. and Maxwell, M. (2015). Probability and Statistics with Applications: A Problem-Solving Text, 2nd ed. ACTEX. ISBN 978-1-62542-472-3.
- Hogg, R. V., Tanis, E. A. and Zimmerman, D. L. (2020). Probability and Statistical Inference, 10th ed. Prentice Hall. ISBN 978-0135189399.

SSSIA-102 Financial Mathematics

SSSIA-102	Financial Mathematics	CREDITS: 4 Total Hours: 60
Type of Course	Theory	15 hours per credit

Course Objectives

The student will understand and be able to:

- (a) perform calculations relating to present value, current value and accumulated value;
- (b) calculate present, current and accumulated value for sequences of non-contingent payments;
- (c) handle loans and bonds and perform the related calculations;
- (d) work with yield curves, rates of return, measures of duration and convexity, cash flow matching and immunisation.

Course Syllabus

At the end of the course, the student will be able to:

Unit	Description	Hours
I	1. Define and recognise the following terms: interest rate, simple interest, compound interest, accumulation function, future value, current value, present value, net present value, discount factor, discount rate, convertible m-thly, nominal rate, effective rate, inflation and real rate of interest, force of interest, equation of value. 2. Given any three of interest rate, period of time, present value and future value, calculate the remaining item using simple or compound interest. Solve time value of money equations involving variable force of interest. 3. Given any one of the effective interest rate, the nominal interest rate convertible m-thly, the effective discount rate, the nominal discount rate convertible m-thly, or the force of interest, calculate any of the other items.	12
II	4. Write the equation of value given a set of cash flows and an interest rate. 5. Define and recognise the following terms: annuity-immediate, annuity-due, perpetuity, payable m-thly or continuously, level payment annuity, arithmetic and geometric increasing or decreasing annuity, term of annuity. 6. For level annuities of finite term, level perpetuities and non-level cash flows in arithmetic or geometric progression, calculate any missing item given sufficient information.	14
III	7. Define and recognise the following terms: principal, interest, term of loan, outstanding balance, final payment including drop and balloon payments, amortisation. 8. Calculate the missing item given any four of term of loan, interest rate, payment amount, payment period and principal. 9. Calculate the outstanding balance at any point in time and the split of a payment between interest and principal repayment. 10. Perform the corresponding calculations where refinancing is involved.	14
IV	11. Define and recognise the following terms: price, book value, market value, amortisation of premium, accumulation of discount, redemption value, par value, yield rate, coupon, coupon rate, term of bond, callable and non-callable, call price, call premium, accumulated value with reinvestment of coupons. 12. Given sufficient partial information, calculate any of the remaining bond items. Valuation of bonds between coupon payment dates is not covered. 13. Construct and interpret a yield curve, including spot and forward rates.	12

V	14. Calculate and interpret net present value, internal rate of return, and dollar-weighted and time-weighted rates of return. 15. Calculate and interpret Macaulay duration, modified duration and convexity. 16. Construct cash flow matching and immunisation strategies for a set of liabilities.	8
	Total	60

References

- Broverman, S. A. (2017). Mathematics of Investment and Credit, 7th ed. ACTEX.
- Vaaler, L. J. F. and Daniel, J. W. (2019). Mathematical Interest Theory, 3rd ed. MAA Press.
- Kellison, S. G. (2008). The Theory of Interest, 3rd ed. McGraw-Hill.

SSSIA-103 Business Economics - Part I (Microeconomics)

SSSIA-103	Business Economics - Part I (Microeconomics)	CREDITS: 3 Total Hours: 45
Type of Course	Theory	15 hours per credit

Course Objectives

The student will understand and be able to:

- (a) key concepts of microeconomics, including the supply and demand curve, market competition and profitability.

Course Syllabus

At the end of the course, the student will be able to:

Unit	Description	Hours
I	1. Explain the concept of utility and how rational utility-maximising agents make consumption choices. 2. Explain consumer surplus and the effect of income and substitution on demand.	11
II	3. Explain the elasticity of supply and demand and the effects on a market of different levels of elasticity. 4. Explain the interaction between supply and demand and the way in which equilibrium market prices are achieved.	12
III	5. Explain the various pricing strategies that can be used by firms. 6. Explain the core economic concepts involved in choices made by businesses with respect to short-run and long-run investment and production choices.	12
IV	7. Explain competitive markets and how they operate. 8. Explain profitability in markets with imperfect competition. 9. Explain asymmetric information, adverse selection and moral hazard as they arise in insurance markets.	10
	Total	45

References

- Sloman, J. (2022). Economics, 11th ed. Pearson. eISBN 978-1292187853.
- Sloman, J., Hinde, K. and Garratt, D. (2019). Economics for Business, 8th ed. Pearson. ISBN 9781292239279.

SSSIA-104 Business Economics - Part II (Macroeconomics)

SSSIA-104	Business Economics - Part II (Macroeconomics)	CREDITS: 3 Total Hours: 45
Type of Course	Theory	15 hours per credit

Course Objectives

The student will understand and be able to:

- (a) key concepts of macroeconomics, including macroeconomic measures, interest rates and exchange rates.

Course Syllabus

At the end of the course, the student will be able to:

Unit	Description	Hours
I	1. Explain the basic macroeconomic measures, such as gross domestic product, used to compare the economies of countries. 2. Describe the structure of public finances for an industrialised country.	11
II	3. Explain the effect of fiscal and monetary policy on the economy, including the effect on financial markets. 4. Explain the role of international trade, exchange rates and the balance of payments in the economy.	12
III	5. Explain the effect of savings and consumption rates on the economy. 6. Explain the major factors affecting the level of interest rates, the rate of inflation, the exchange rate, the level of employment and the rate of growth for an industrialised country.	12
IV	7. Describe the function of money in the economy and explain the relationship between money and interest rates. 8. Explain how macroeconomic policies affect businesses, including general insurance undertakings.	10
	Total	45

References

- Sloman, J. (2022). Economics, 11th ed. Pearson. eISBN 978-1292187853.
- Sloman, J., Hinde, K. and Garratt, D. (2019). Economics for Business, 8th ed. Pearson. ISBN 9781292239279.

SSSIA-105 Corporate Finance

SSSIA-105	Corporate Finance	CREDITS: 3 Total Hours: 45
Type of Course	Theory	15 hours per credit

Course Objectives

The student will understand and be able to:

- (a) key concepts of corporate finance, including capital structure, the different financial instruments and their characteristics.

Course Syllabus

At the end of the course, the student will be able to:

Unit	Description	Hours
I	1. Definitions of key finance terms: stock company and capital structure. 2. Key finance concepts: financing companies, characteristics and uses of financial instruments, sources of capital, cost of capital, dividend policy, personal and corporate taxation. 3. Factors to be considered by a company when deciding on its capital structure and dividend policy.	11
II	4. The impact of financial leverage and of long-term and short-term financing policies on capital structure. 5. Characteristics of the principal forms of financial instruments issued or used by companies, and the ways in which they may be issued. 6. How a company's cost of capital relates to the investment projects the company wishes to undertake.	12
III	7. Definitions of key finance terms: bonds, stock, basic options including calls and puts, dividends and the price to earnings ratio. 8. The structure of a stock company and the different methods by which it may be financed. 9. Calculate the value of stocks.	12
IV	10. Measures of financial performance: balance sheet, income statement, statement of cash flows and financial ratios covering leverage, liquidity, profitability and market value. 11. Assessment of financial performance using net present value, payback and discounted payback models, internal rate of return and the profitability index.	10
	Total	45

References

- Ross, S., Westerfield, R. and Jaffe, J. (2002). Corporate Finance, 6th ed. McGraw-Hill.
- Berk, J. and DeMarzo, P. (2007). Corporate Finance. Pearson International.
- Brealey, R. A., Myers, S. C. and Allen, F. (2003). Principles of Corporate Finance, 7th ed. McGraw-Hill.
- Copeland, T., Weston, F. and Shastri, K. (2004). Financial Theory and Corporate Policy, 4th ed. Addison-Wesley.

SSSIA-106 Financial and Insurance Accounting

SSSIA-106	Financial and Insurance Accounting	CREDITS: 3 Total Hours: 45
Type of Course	Theory	15 hours per credit

Course Objectives

The student will understand and be able to:

- (a) key concepts of corporate accounting and finance, including taxation and capital budgeting, and their application to insurance undertakings.

Course Syllabus

At the end of the course, the student will be able to:

Unit	Description	Hours
I	1. Describe the basic principles of corporate taxation and the taxation of investments held by institutions. 2. Explain why companies are required to produce annual reports. 3. Explain fundamental accounting concepts and terms, and describe the main sources of accounting regulation.	11
II	4. Explain the structure and purpose of the income statement, balance sheet and statement of cash flows, and the interactions between them. 5. Construct simple statements of financial position and profit or loss. 6. Calculate and interpret financial ratios.	12
III	7. Explain the characteristics of the various forms of equity capital from the point of view of the issuer and the investor. 8. Explain the characteristics of the various forms of long-term debt capital from the point of view of the issuer and the investor. 9. Explain the characteristics of the various forms of short-term and medium-term financing.	12
IV	10. Calculate the weighted average cost of capital. 11. Explain the main methods of capital budgeting and calculate a project's investment return. 12. Describe the accounting treatment of premium earning patterns, unearned premium, claims liabilities and reinsurance in insurance company accounts.	10
	Total	45

References

- Atrill, P. and McLaney, E. (2015). Management Accounting for Decision Makers, 9th ed. Prentice Hall. ISBN 9781292062716.
- Berry, A. and Jarvis, R. (2011). Corporate Finance: Principles and Practice, 5th ed. Cengage. ISBN 9781408030479.
- Leiwy, D. and Perks, R. (2015). Accounting: Understanding and Practice, 4th ed. McGraw-Hill. ISBN 9780077139131.

SSSIA-107 Modern Actuarial Statistics I - Part I

SSSIA-107	Modern Actuarial Statistics I - Part I	CREDITS: 4 Total Hours: 60
Type of Course	Theory	15 hours per credit

Course Objectives

The student will understand and be able to:

- (a) solve problems using stochastic processes and determine the probabilities and distributions associated with these processes;
- (b) perform survival model and life contingency calculations;
- (c) estimate parameters and test statistical hypotheses on insurance data.

Course Syllabus

At the end of the course, the student will be able to:

Unit	Description	Hours
I	1. Model claim frequencies using Poisson processes. 2. Calculate expected values, variances and probabilities for any Poisson process. 3. Calculate the limited expected value.	12
II	4. Perform survival model and hazard rate calculations. 5. Perform joint life calculations. 6. Calculate simple whole life or annuity problems.	14
III	7. Estimate the mean and variance given a sample and estimate a sufficient statistic for a distribution. 8. Test statistical hypotheses, including Type I and Type II errors. 9. Test means and variances using critical values from a sampling distribution.	12
IV	10. Model insurance claim frequency and severity. 11. Model insurance claims in aggregate. 12. Calculate order statistics of a sample.	14
V	13. Perform point estimation of statistical parameters using maximum likelihood estimation, applying criteria such as consistency, unbiasedness, sufficiency, efficiency, minimum variance and mean square error. 14. Adjust calculations for the effect of missing data values, including censoring and truncation.	8
	Total	60

References

- Daniel, J. W. (2008). Poisson Processes (and Mixture Distributions). CAS Study Note.
- Ross, S. M. (2019). Introduction to Probability Models, 12th ed. Academic Press.
- Struppeck, T. (2015). Life Contingencies. CAS Study Note.
- Hogg, R. V., McKean, J. W. and Craig, A. T. (2018). Introduction to Mathematical Statistics, 8th ed. Prentice Hall.
- Tse, Y. (2009). Nonlife Actuarial Models: Theory, Methods and Evaluation. Cambridge University Press.

SSSIA-108 Modern Actuarial Statistics I - Part II

SSSIA-108	Modern Actuarial Statistics I - Part II	CREDITS: 4 Total Hours: 60
Type of Course	Theory	15 hours per credit

Course Objectives

The student will understand and be able to:

- (a) solve problems using extended linear models and determine when these models are appropriate to use;
- (b) evaluate, interpret and diagnose the output of an extended linear model.

Course Syllabus

At the end of the course, the student will be able to:

Unit	Description	Hours
I	1. Select the appropriate model for an extended linear model. 2. Select the appropriate model structure given the behaviour of the data set to be modelled, including the appropriate link function and distribution for the dependent variable in a generalised linear model.	12
II	3. Evaluate models developed using an extended linear model approach. 4. Interpret the extended linear model output from statistical software, such as parameter estimate tables and analysis of variance tables.	14
III	5. Distinguish among categorical, ordinal and continuous predictors and their interactions, and how these relate to their usage in an extended linear model. 6. Understand and apply control and offset variables in generalised linear models.	12
IV	7. Understand and calculate the Akaike information criterion, the Bayesian information criterion, deviance and R-squared. 8. Analyse model diagnostic plots, including residual, marginal model, quantile-quantile and added variable plots, to assess quality of fit.	14
V	9. Interpret exploratory data analysis plots for various data types, including box plots, univariate plots and histograms.	8
	Total	60

References

- Dobson, A. and Barnett, A. (2018). An Introduction to Generalized Linear Models, 4th ed. Chapman and Hall/CRC.
- Hogg, R. V., McKean, J. W. and Craig, A. T. (2018). Introduction to Mathematical Statistics, 8th ed. Prentice Hall.
- James, G. et al. (2021). An Introduction to Statistical Learning with Applications in R, 2nd ed. Springer.
- Larsen, M. (2016). Generalized Linear Models. CAS Study Note.

SSSIA-109 Modern Actuarial Statistics II - Part I

SSSIA-109	Modern Actuarial Statistics II - Part I	CREDITS: 4 Total Hours: 60
Type of Course	Theory	15 hours per credit

Course Objectives

The student will understand and be able to:

- (a) calculate credibility weighted estimates;
- (b) explain the structure of linear mixed models, including models with correlated observations and models where the variance is not constant or is a function of the mean;
- (c) describe the basic applications of the autoregressive integrated moving average time series model.

Course Syllabus

At the end of the course, the student will be able to:

Unit	Description	Hours
I	1. Calculate classical limited fluctuation, Buhlmann, Buhlmann-Straub and Bayesian credibility weighted estimates for frequency, severity and aggregate loss. 2. Understand the framework used for each of these credibility procedures.	12
II	3. Understand the assumptions behind the linear mixed model design. 4. Understand how to use a hierarchical model.	14
III	5. Interpret output from a linear mixed model and make appropriate choices when evaluating modelling options. 6. Interpret linear mixed model diagnostics and summary statistics to evaluate the model structure and variable selection.	12
IV	7. Model relationships of current and past values of a statistic or metric. 8. Understand the framework of ARIMA models, including trends and seasonality.	14
V	9. Calculate trends and seasonality using time series with regression, distinguishing deterministic from stochastic trend. 10. Interpret time series output to make forecasts.	8
	Total	60

References

- Tse, Y. (2009). Nonlife Actuarial Models: Theory, Methods and Evaluation. Cambridge University Press.
- West, B. T., Welch, K. B. and Galecki, A. T. (2022). Linear Mixed Models: A Practical Guide Using Statistical Software, 3rd ed. CRC Press.
- Cowpertwait, P. and Metcalfe, A. (2009). Introductory Time Series with R. Springer.

SSSIA-110 Modern Actuarial Statistics II - Part II

SSSIA-110	Modern Actuarial Statistics II - Part II	CREDITS: 4 Total Hours: 60
Type of Course	Theory	15 hours per credit

Course Objectives

The student will understand and be able to:

- (a) understand the mechanics of the statistical learning algorithms listed below and recognise their inherent strengths and weaknesses, so as to select the most appropriate procedure for the learning task at hand.

Course Syllabus

At the end of the course, the student will be able to:

Unit	Description	Hours
I	1. Compute K-nearest neighbours. 2. Prune decision trees.	12
II	3. Calculate the summary statistics for a set of decision trees. 4. Understand the assumptions underlying different tree ensemble methods and the improvements they can make to decision trees.	14
III	5. Compute the elements of principal components analysis, including loading vectors and variance explained. 6. Interpret principal components analysis software output.	12
IV	7. Perform the computations behind clustering procedures, including K-means and hierarchical clustering, and interpret their output. 8. Interpret neural network results.	14
V	9. Calculate measures of model predictive accuracy, including entropy, Gini, the confusion matrix and lift. 10. Compare models via predictive performance measures such as the double lift chart.	8
	Total	60

References

- James, G. et al. (2021). An Introduction to Statistical Learning with Applications in R, 2nd ed. Springer.
- Goldburd, M. et al. (2020). Generalized Linear Models for Insurance Rating, 2nd ed. CAS Monograph No. 5.

SSSIA-111 Ratemaking - Part I

SSSIA-111	Ratemaking - Part I	CREDITS: 4 Total Hours: 60
Type of Course	Theory	15 hours per credit

Course Objectives

The student will understand and be able to:

- (a) analyse ratemaking data, select appropriate techniques and develop solutions to ratemaking problems;
- (b) assemble, organise and adjust the exposure, premium and loss data required for a rate level indication.

Course Syllabus

At the end of the course, the student will be able to:

Unit	Description	Hours
I	1. Define and describe exposure bases as used in the ratemaking process. 2. Evaluate and select an exposure base in a given scenario, having regard to the line of business and use case.	12
II	3. Aggregate and organise ratemaking data by calendar year, policy year, accident year, report year, close year, in force and net of reinsurance. 4. Evaluate ratemaking data and analyses for errors and reasonableness.	12
III	5. Separate data into groups that balance homogeneity and credibility, and summarise the considerations for determining such groups. 6. Calculate loss and loss adjustment expenses to be used for ratemaking, including common ratios, adjustments to losses and the distinction between claims-made and occurrence coverage.	12
IV	7. Calculate the various adjustments to premium used for ratemaking, including on-levelling and premium audit. 8. Calculate and apply exposure, premium and loss trends using different approaches, including exponential and linear analyses.	14
V	9. Calculate the underwriting provisions underlying the overall rate level indication, including fixed and variable expenses, profit and contingency loading and reinsurance costs. 10. Demonstrate proper handling of extraordinary losses for ratemaking purposes, including large losses and catastrophes.	10
	Total	60

References

- Werner, G. and Modlin, C. (2016). Basic Ratemaking, 5th ed. Casualty Actuarial Society.
- Casualty Actuarial Society (1988). Statement of Principles Regarding Property and Casualty Insurance Ratemaking.
- Actuarial Standards Board. Actuarial Standard of Practice No. 13, Trending Procedures in Property/Casualty Insurance.

SSSIA-112 Ratemaking - Part II

SSSIA-112	Ratemaking - Part II	CREDITS: 4 Total Hours: 60
Type of Course	Theory	15 hours per credit

Course Objectives

The student will understand and be able to:

- (a) construct an overall rate level indication and translate it into an implemented rating structure;
- (b) perform classification, territorial and individual risk rating calculations;
- (c) apply the principles of ratemaking and the constraints that govern implementation.

Course Syllabus

At the end of the course, the student will be able to:

Unit	Description	Hours
I	1. Construct an overall rate level indication using the pure premium and loss ratio methods and reconcile the two. 2. Apply credibility to ratemaking analyses using different methods and justify the choice of complement of credibility.	14
II	3. Perform calculations related to alternative ratemaking procedures, including classification and territorial ratemaking. 4. Perform calculations for deductibles, increased limits and coinsurance.	12
III	5. Perform calculations for commercial lines rating mechanisms, including schedule rating, experience rating and retrospective rating. 6. Analyse the results of predictive models, including generalised linear models, in a ratemaking context.	12
IV	7. Select and justify a final rate change to implement beyond the calculated overall rate level indication, having regard to operational, marketing and regulatory constraints and to lifetime value. 8. Apply the four principles of ratemaking to a scenario.	12
V	9. Understand the considerations for implementing rates to achieve an organisation's goals, including non-pricing solutions, minimum premium and rating algorithms. 10. Perform dislocation analysis and apply capping strategies on implementation.	10
	Total	60

References

- Werner, G. and Modlin, C. (2016). Basic Ratemaking, 5th ed. Casualty Actuarial Society.
- Actuarial Standards Board. Actuarial Standard of Practice No. 12, Risk Classification.
- Goldburd, M. et al. (2020). Generalized Linear Models for Insurance Rating, 2nd ed. CAS Monograph No. 5.

SSSIA-113 Reserving - Part I

SSSIA-113	Reserving - Part I	CREDITS: 4 Total Hours: 60
Type of Course	Theory	15 hours per credit

Course Objectives

The student will understand and be able to:

- (a) estimate unpaid claims for insurance entities and for non-insurance entities that retain risk;
- (b) organise and diagnose reserving data and build development triangles;
- (c) apply the standard deterministic estimation techniques.

Course Syllabus

At the end of the course, the student will be able to:

Unit	Description	Hours
I	1. Organise reserving data by calendar year, accident year, policy year, underwriting year and report year. 2. Evaluate reserving data and analyses for errors and reasonableness. 3. Describe the role of homogeneity and credibility of data in the process of estimating unpaid claims.	12
II	4. Describe the fundamentals of different types of insurance, including long tail versus short tail lines, low frequency versus high frequency lines, and occurrence versus claims-made coverage. 5. Articulate the importance of accurate estimates of unpaid claims.	12
III	6. Build and analyse development triangles for losses, claim counts and allocated loss adjustment expenses. 7. Apply a tail factor.	12
IV	8. Use development triangles as diagnostic tools to identify changes and trends, including loss and claim count patterns, ratios of losses to premium, severity and frequency ratios. 9. Calculate and evaluate the development or chain ladder method and the case outstanding development method.	14
V	10. Calculate and evaluate the expected loss, Bornhuetter-Ferguson, Cape Cod, Benktander and frequency-severity methods. 11. Calculate and evaluate the Berquist-Sherman adjustments.	10
	Total	60

References

- Friedland, J. F. (2010). Estimating Unpaid Claims Using Basic Techniques, 3rd version. Casualty Actuarial Society.
- Actuarial Standards Board. Actuarial Standard of Practice No. 43, Property/Casualty Unpaid Claim Estimates.

SSSIA-114 Reserving - Part II

SSSIA-114	Reserving - Part II	CREDITS: 4 Total Hours: 60
Type of Course	Theory	15 hours per credit

Course Objectives

The student will understand and be able to:

- (a) adjust reserving data and techniques for internal and external change;
- (b) estimate recoveries and loss adjustment expense liabilities;
- (c) evaluate, monitor and communicate the results of a reserve analysis.

Course Syllabus

At the end of the course, the student will be able to:

Unit	Description	Hours
I	1. Assess the influence of operating changes on the estimation of unpaid losses, including claims coding, claims processing, underwriting and policy provisions, marketing, reinsurance and the treatment of recoveries.	12
II	2. Adjust data and estimation techniques for changes in the internal and external environment, including shifts in case outstanding adequacy, shifts in settlement rates, changes in mix of business, changes in rate level and changes in the inflationary or legal environment. 3. Consider the impact of, and adjust for, the presence of large losses in a reserving analysis.	14
III	4. Calculate and evaluate estimation techniques for recoveries, including salvage, subrogation and reinsurance. 5. Define and apply reinsurance concepts to calculate net, ceded and gross losses.	12
IV	6. Calculate and evaluate estimation techniques for allocated loss adjustment expenses. 7. Calculate and evaluate estimation techniques for unallocated loss adjustment expenses.	12
V	8. Evaluate the results of a reserve analysis for adequacy and reasonableness using loss ratios, severities, pure premiums, frequencies and indicated unpaid losses. 9. Monitor results for adequacy and reasonability, including interim valuations, actual versus expected and roll forward analysis. 10. Utilise external information in a reserve analysis and communicate results and drivers of change to internal management, investors and regulators.	10
	Total	60

References

- Friedland, J. F. (2010). Estimating Unpaid Claims Using Basic Techniques, 3rd version. Casualty Actuarial Society.
- Actuarial Standards Board. Actuarial Standard of Practice No. 43, Property/Casualty Unpaid Claim Estimates.

SSSIA-115 Regulation and Financial Reporting - Part I

SSSIA-115	Regulation and Financial Reporting - Part I	CREDITS: 5 Total Hours: 75
Type of Course	Theory	15 hours per credit

Course Objectives

The student will understand and be able to:

- (a) understand the role of regulators in requiring insurers to demonstrate that they are providing fair and reliable services in accordance with the statutes and regulations of each jurisdiction;
- (b) understand marketplace function, marketplace conduct and the government and industry programmes that operate alongside private insurance.

Course Syllabus

At the end of the course, the student will be able to:

Unit	Description	Hours
I	1. Understand marketplace regulation and the rationale for regulating insurance. 2. Describe the objectives of regulation: solvency, market conduct and policyholder protection.	15
II	3. Understand marketplace function, including the roles of insurers, intermediaries and reinsurers. 4. Understand rate regulation and policy form approval.	15
III	5. Understand marketplace conduct and the issues arising, including discrimination in insurance and fair treatment of customers.	15
IV	6. Understand mainstream government and industry programmes and residual market mechanisms. 7. Understand the anatomy of an insurance policy and the analysis of coverage.	15
V	8. Understand insurance accounting and the reporting entity's statutory returns. 9. Understand the Indian regulatory framework and the role of the IRDAI, with comparison to selected international supervisory regimes.	15
	Total	75

References

- Brown, E. F. and Klein, R. W. (2015). Research Handbook on the Law and Economics of Insurance. Edward Elgar Publishing.
- Avraham, R. (2017). Discrimination and Insurance. The Routledge Handbook to Discrimination.
- Eling, M. (2021). Insurance Regulation in Europe: An Analysis of Effectiveness and Efficiency. Journal of Insurance Regulation, 40(3), NAIC.
- International Association of Insurance Supervisors. Insurance Core Principles (current edition).

SSSIA-116 Regulation and Financial Reporting - Part II

SSSIA-116	Regulation and Financial Reporting - Part II	CREDITS: 5 Total Hours: 75
Type of Course	Theory	15 hours per credit

Course Objectives

The student will understand and be able to:

- (a) understand the concept and assessment of solvency, including the Own Risk and Solvency Assessment and the various international approaches to assessing solvency;
- (b) apply international financial reporting requirements to general insurance contracts.

Course Syllabus

At the end of the course, the student will be able to:

Unit	Description	Hours
I	1. Understand the components of solvency and the purpose of solvency regulation. 2. Understand technical provisions, best estimate liabilities and the risk margin.	15
II	3. Understand the capital adequacy components of solvency regulation, including risk-based capital. 4. Understand the calculation and interpretation of solvency capital requirements.	15
III	5. Understand the Own Risk and Solvency Assessment and enterprise risk management in general. 6. Understand the board's perspective on deriving value from ORSA.	15
IV	7. Understand mainstream global solvency assessment frameworks, including Solvency II, the Bermuda and Swiss approaches and the Indian solvency framework.	15
V	8. Apply the IFRS 17 general measurement model and the premium allocation approach to general insurance contracts, including fulfilment cash flows, risk adjustment, discounting and the contractual service margin. 9. Understand presentation, disclosure and reinsurance held under IFRS 17, and the role of the reporting actuary.	15
	Total	75

References

- Blanchard, R. S. (2021). Exam 6-International Study Note: Solvency. CAS Study Note.
- International Actuarial Association, Insurer Solvency Assessment Working Party (2004). A Global Framework for Insurer Solvency Assessment.
- International Actuarial Association (2016). IAA Risk Book.
- International Actuarial Association (2015). Deriving Value from ORSA: Board Perspective.
- International Accounting Standards Board. IFRS 17 Insurance Contracts.

SSSIA-117 Predictive Analytics - Part I

SSSIA-117	Predictive Analytics - Part I	CREDITS: 6 Total Hours: 90
Type of Course	Theory	15 hours per credit

Course Objectives

The student will understand and be able to:

- (a) identify the business problem, relate the available data to possible analyses and propose an appropriate modelling approach;
- (b) work with various data types, understand the principles of data design and construct common visualisations for exploring data;
- (c) understand the key concepts of statistical learning and of generalised linear models.

Course Syllabus

At the end of the course, the student will be able to:

Unit	Description	Hours
I	1. Assess whether descriptive, predictive or prescriptive analytics applies to a business problem. 2. Describe the characteristics of predictive modelling problems. 3. Translate a vague question into one that can be analysed with statistics and predictive analytics to solve a business problem.	18
II	4. Define the terms used to classify types of modelling problems and methods, including supervised versus unsupervised learning and regression versus classification. 5. Compare the common methods of assessing model accuracy. 6. Explain the concepts of bias, variance, model complexity and the bias-variance trade-off.	18
III	7. Identify structured and unstructured data types and the types of variables used in predictive modelling. 8. Evaluate effective data design with respect to time frame, sampling and granularity. 9. Apply the key principles of constructing graphs and apply univariate and bivariate data exploration techniques.	18
IV	10. Understand resampling methods used for model validation, including the training and test set approach, k-fold cross-validation and leave-one-out cross-validation. 11. Compare model assumptions for ordinary least squares and generalised linear models. 12. Identify the members of the exponential family of distributions and the corresponding link functions.	18
V	13. Apply the business context of a problem to interpret parameters. 14. Interpret diagnostic tests of model fit and assumption checking using graphical and quantitative methods. 15. Select an appropriate model considering distributions and link functions, variable transformations and interactions, t and F tests, AIC, BIC and the likelihood ratio test. 16. Calculate and interpret predicted values and confidence and prediction intervals.	18
	Total	90

References

- Frees, E. W. (2010). Regression Modeling with Actuarial and Financial Applications. Cambridge University Press. ISBN 978-0521135962.

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- James, G., Witten, D., Hastie, T. and Tibshirani, R. (2021). An Introduction to Statistical Learning with Applications in R, 2nd ed. Springer.
 - Healy, K. (2018). Data Visualization: A Practical Introduction. Princeton University Press. ISBN 978-0-691-18162-2.

SSSIA-118 Predictive Analytics - Part II

SSSIA-118	Predictive Analytics - Part II	CREDITS: 6 Total Hours: 90
Type of Course	Theory	15 hours per credit

Course Objectives

The student will understand and be able to:

- (a) transform data, including by the application of unsupervised learning techniques, to improve predictive modelling outcomes;
- (b) apply regression-based time series models, decision tree models, principal component analysis and cluster analysis;
- (c) select, validate and communicate a model appropriate to a given insurance business problem.

Course Syllabus

At the end of the course, the student will be able to:

Unit	Description	Hours
I	1. Create features from existing data that may add value. 2. Apply principal components analysis to transform data and interpret loading factors and proportion of variance explained. 3. Apply K-means and hierarchical clustering to transform data and explain methods for deciding the number of clusters.	18
II	4. Define the concepts and components of stochastic time series processes, including random walks, stationarity and autocorrelation. 5. Describe specific time series models, including exponential smoothing, autoregressive and autoregressive conditionally heteroskedastic models. 6. Calculate and interpret predicted values and confidence intervals.	18
III	7. Describe the construction of decision trees, including how they are optimally fit to training data and how they are pruned to mitigate overfitting. 8. Predict outcomes using classification trees and regression trees. 9. Compare decision trees to linear models, including their uses and relative strengths.	18
IV	10. Describe bagging, boosting and random forests and the hyperparameters used to control them. 11. Select appropriate hyperparameters for regularised regression and apply lasso and ridge regression. 12. Apply offsets and weights as appropriate and interpret model coefficients, including interaction terms.	18
V	13. Select and validate a generalised linear model as appropriate for a business problem. 14. Evaluate model performance using lift charts, double lift charts, gains curves and the confusion matrix. 15. Communicate model construction, limitations and business implications to a non-technical audience, having regard to regulatory and fairness considerations.	18
	Total	90

References

- Frees, E. W. (2010). Regression Modeling with Actuarial and Financial Applications. Cambridge University Press.

- James, G., Witten, D., Hastie, T. and Tibshirani, R. (2021). An Introduction to Statistical Learning with Applications in R, 2nd ed. Springer.
- Goldburd, M. et al. (2020). Generalized Linear Models for Insurance Rating, 2nd ed. CAS Monograph No. 5.
- Lander, J. (2017). R for Everyone, 2nd ed. Addison-Wesley. ISBN 978-0-13-454692-6.

SSSIA-119 Predictive Analytics Project Practicum

SSSIA-119	Predictive Analytics Project Practicum	CREDITS: 4 Total Hours: 120
Type of Course	Practical	30 hours per credit

Course Objectives

The student will understand and be able to:

- (a) execute an end-to-end predictive modelling project independently within a fixed working window;
- (b) document every modelling decision and its justification;
- (c) produce a written report to professional standard and defend it under questioning.

Course Syllabus

At the end of the course, the student will be able to:

Unit	Description	Hours
I	1. Project brief, data dictionary and scoping; agreeing the business objective in writing. 2. Project planning within a fixed working window.	20
II	3. Data validation, cleaning and preparation with a documented audit trail. 4. Exploratory analysis and the record of approaches considered and rejected.	28
III	5. Model construction, hyperparameter tuning and validation. 6. Sensitivity and stability testing.	30
IV	7. Report writing: structure, exhibits, assumptions, limitations and recommendations. 8. Communication of results to a non-technical stakeholder.	26
V	9. Viva voce defence of the submitted project. 10. Peer review and reflective practice.	16
	Total	120

References

- Actuarial Standards Board. Actuarial Standard of Practice No. 56, Modeling.
- Kuhn, M. and Johnson, K. (2013). Applied Predictive Modeling. Springer.
- Goldburd, M. et al. (2020). Generalized Linear Models for Insurance Rating, 2nd ed. CAS Monograph No. 5.

SSSIA-120 Agentic AI for Probability

SSSIA-120	Agentic AI for Probability	CREDITS: 3 Total Hours: 90
Type of Course	Practical	30 hours per credit

Course Objectives

The student will understand and be able to:

- (a) use large language model agents to set up, check and explain probability calculations;
- (b) build a tool-using agent that calls a numerical or simulation backend rather than reasoning numerically on its own;
- (c) verify agent output against an independent calculation and recognise the failure modes of generated reasoning.

Course Syllabus

At the end of the course, the student will be able to:

Unit	Description	Hours
I	1. Anatomy of an agent: model, system prompt, tools, memory and the control loop. Comparison with a single prompt-response interaction. 2. Setting up a local agent environment and connecting a Python execution tool.	16
II	3. Structuring a probability problem for an agent: stating the sample space, the random variables and the quantity sought in machine-readable form. 4. Delegating distribution fitting and moment calculation to a computation tool and reading back the result.	20
III	5. Building a Monte Carlo simulation agent for loss random variables under deductibles, limits and coinsurance. 6. Comparing simulated results against closed-form answers and quantifying simulation error.	20
IV	7. Verification and guardrails: independent recomputation, unit and sanity checks, and refusal behaviour when the agent lacks information. 8. Cataloguing observed failure modes, including confident arithmetic errors and misidentified distributions.	18
V	9. Building an explanation agent that produces a worked solution for a probability problem at a stated level of detail. 10. Evaluating generated explanations for correctness and pedagogical quality.	16
	Total	90

References

- Anthropic. Building Effective Agents (engineering guidance, current edition).
- Model Context Protocol specification and reference servers.
- Ross, S. M. (2019). A First Course in Probability, 10th ed. Pearson (for verification problems).
- Hassett, M., Stewart, D. and Milovanovic, J. (2021). Probability for Risk Management, 3rd ed. ACTEX.

SSSIA-121 Agentic AI for Financial Mathematics

SSSIA-121	Agentic AI for Financial Mathematics	CREDITS: 3 Total Hours: 90
Type of Course	Practical	30 hours per credit

Course Objectives

The student will understand and be able to:

- (a) automate valuation, amortisation and immunisation calculations through tool-using agents;
- (b) build an agent that constructs and audits a cash flow model from a natural-language specification;
- (c) assess where agent automation is and is not appropriate in a financial calculation workflow.

Course Syllabus

At the end of the course, the student will be able to:

Unit	Description	Hours
I	1. Representing cash flows, term structures and equations of value in a form an agent can manipulate. 2. Connecting an agent to a spreadsheet or a financial library as a tool.	16
II	3. An agent that builds an amortisation schedule from a natural-language loan description and explains each column. 4. Handling refinancing, drop and balloon payments through agent-driven recalculation.	20
III	5. An agent that prices a bond, produces the premium amortisation or discount accumulation schedule, and flags inconsistent inputs. 6. Yield curve construction and bootstrapping of spot rates under agent control.	20
IV	7. Duration, convexity and immunisation: an agent that proposes a matching portfolio and tests it under interest rate shocks. 8. Scenario generation and comparison of agent-proposed strategies.	18
V	9. Auditing agent-built models: reconstructing the calculation independently, tracing tool calls and documenting the audit trail. 10. Deciding what to automate and what to retain under human calculation.	16
	Total	90

References

- Broverman, S. A. (2017). Mathematics of Investment and Credit, 7th ed. ACTEX (for verification problems).
- Anthropic. Building Effective Agents (engineering guidance, current edition).
- Model Context Protocol specification and reference servers.

SSSIA-122 Agentic AI for Modern Actuarial Statistics I

SSSIA-122	Agentic AI for Modern Actuarial Statistics I	CREDITS: 3 Total Hours: 90
Type of Course	Practical	30 hours per credit

Course Objectives

The student will understand and be able to:

- (a) use agents to assist with model specification, fitting and diagnosis for generalised linear models and survival models;
- (b) build an agent workflow that documents every modelling decision as it is taken;
- (c) critically evaluate agent-proposed model structures against statistical judgement.

Course Syllabus

At the end of the course, the student will be able to:

Unit	Description	Hours
I	1. Connecting an agent to a statistical environment; passing data summaries rather than raw data and the reasons for doing so. 2. Prompt and tool design for reproducible statistical work.	16
II	3. An agent that proposes candidate distributions and link functions for a claim frequency or severity data set and justifies each proposal. 4. Automated fitting, with the analyst retaining the selection decision.	20
III	5. Agent-generated diagnostics: residual and quantile-quantile analysis, deviance, AIC and BIC comparison, and narrative interpretation of each. 6. Detecting and challenging incorrect agent interpretations of diagnostic output.	20
IV	7. Survival and hazard rate estimation with censored and truncated data under agent assistance. 8. Checking agent handling of censoring, a frequent source of silent error.	18
V	9. Automatic generation of a model documentation pack from the agent's tool call history. 10. Standards for disclosing the use of AI assistance in actuarial work.	16
	Total	90

References

- Dobson, A. and Barnett, A. (2018). An Introduction to Generalized Linear Models, 4th ed. Chapman and Hall/CRC.
- Actuarial Standards Board. Actuarial Standard of Practice No. 56, Modeling.
- Anthropic. Building Effective Agents (engineering guidance, current edition).

SSSIA-123 Agentic AI for Modern Actuarial Statistics II

SSSIA-123	Agentic AI for Modern Actuarial Statistics II	CREDITS: 3 Total Hours: 90
Type of Course	Practical	30 hours per credit

Course Objectives

The student will understand and be able to:

- (a) orchestrate multi-step statistical learning workflows through agents;
- (b) use agents for credibility, mixed model and time series work with documented parameter choices;
- (c) evaluate the reproducibility and stability of agent-driven analyses.

Course Syllabus

At the end of the course, the student will be able to:

Unit	Description	Hours
I	1. Multi-step agent workflows: planning, decomposition, delegation to sub-agents and result aggregation. 2. State and memory management across a long analysis.	16
II	3. Agent-assisted credibility work: parameter estimation for Buhlmann and Buhlmann-Straub models with the assumptions surfaced explicitly. 4. Testing whether the agent has correctly identified the credibility framework in use.	20
III	5. Linear mixed models and hierarchical structures under agent assistance; interpretation of variance components. 6. Time series model identification, fitting and forecasting with agent support.	20
IV	7. Hyperparameter search and ensemble construction driven by an agent, with a cost and compute budget. 8. Reproducibility: seeding, environment capture and re-running an agent analysis to the same result.	18
V	9. Stability testing: re-running the same brief and comparing the agent's choices across runs. 10. Reporting variability in agent-driven model selection as a limitation.	16
	Total	90

References

- James, G. et al. (2021). An Introduction to Statistical Learning with Applications in R, 2nd ed. Springer.
- West, B. T., Welch, K. B. and Galecki, A. T. (2022). Linear Mixed Models: A Practical Guide Using Statistical Software, 3rd ed. CRC Press.
- Anthropic. Building Effective Agents (engineering guidance, current edition).

SSSIA-124 Agentic AI for Ratemaking

SSSIA-124	Agentic AI for Ratemaking	CREDITS: 4 Total Hours: 120
Type of Course	Practical	30 hours per credit

Course Objectives

The student will understand and be able to:

- (a) build an agent-assisted pipeline from raw policy and claims extracts to a rate level indication;
- (b) automate the repetitive elements of a ratemaking analysis while retaining actuarial control of every selection;
- (c) produce a rate filing exhibit pack and supporting narrative from an agent workflow.

Course Syllabus

At the end of the course, the student will be able to:

Unit	Description	Hours
I	1. Designing a ratemaking agent pipeline: the stages, the tools at each stage and the points at which a human selection is mandatory. 2. Data ingestion agents for policy, premium and claims extracts, with schema validation.	22
II	3. Agent-driven premium on-levelling: extension of exposures and the parallelogram method, with the rate change history supplied as structured data. 4. Automated trend selection candidates, with the agent presenting options and the analyst selecting.	26
III	5. Agent-assisted loss development and large loss identification for a ratemaking data set. 6. Expense provision assembly and the permissible loss ratio, generated and cross-checked.	26
IV	7. Overall rate level indication produced by both the pure premium and the loss ratio method, with the agent reconciling the two and explaining any difference. 8. Classification relativity analysis with agent-supported generalised linear modelling.	26
V	9. Exhibit generation: assembling a filing-ready pack with the agent, and verifying every number against source. 10. Drafting the supporting narrative and disclosure of AI assistance in a regulatory submission.	20
	Total	120

References

- Werner, G. and Modlin, C. (2016). Basic Ratemaking, 5th ed. Casualty Actuarial Society.
- Goldburd, M. et al. (2020). Generalized Linear Models for Insurance Rating, 2nd ed. CAS Monograph No. 5.
- Actuarial Standards Board. Actuarial Standard of Practice No. 56, Modeling.
- Anthropic. Building Effective Agents (engineering guidance, current edition).

SSSIA-125 Agentic AI for Reserving

SSSIA-125	Agentic AI for Reserving	CREDITS: 4 Total Hours: 120
Type of Course	Practical	30 hours per credit

Course Objectives

The student will understand and be able to:

- (a) automate triangle construction, diagnostic production and method application in a reserving analysis;
- (b) use agents to surface data anomalies and environmental changes that a manual review would miss;
- (c) generate a reserve report and actual-versus-expected monitoring pack under agent assistance.

Course Syllabus

At the end of the course, the student will be able to:

Unit	Description	Hours
I	1. Reserving data pipelines: transaction-level extracts to development triangles by accident, policy and report year, built and validated by an agent. 2. Automated data quality testing and reconciliation to the financial statements.	22
II	3. Agent-generated diagnostic suites: link ratio patterns, paid to incurred ratios, frequency and severity trends, and closure rate analysis. 4. Prompting the agent to identify and articulate what the diagnostics suggest about operational change.	26
III	5. Method application across development, expected loss, Bornhuetter-Ferguson, Cape Cod and Benktander approaches under agent orchestration. 6. Agent-supported Berquist-Sherman adjustment where case adequacy or settlement rates have shifted.	26
IV	7. Selection: the agent presents method results and rationale, the analyst selects, and the selection reasoning is recorded automatically. 8. Tail factor fitting and sensitivity analysis.	26
V	9. Report generation: exhibits, commentary and drivers of change, assembled by the agent and verified line by line. 10. Actual versus expected and roll forward monitoring as a scheduled agent task.	20
	Total	120

References

- Friedland, J. F. (2010). Estimating Unpaid Claims Using Basic Techniques, 3rd version. Casualty Actuarial Society.
- Actuarial Standards Board. Actuarial Standard of Practice No. 43, Property/Casualty Unpaid Claim Estimates.
- Actuarial Standards Board. Actuarial Standard of Practice No. 56, Modeling.
- Anthropic. Building Effective Agents (engineering guidance, current edition).

SSSIA-126 Agentic AI for Regulation and Reporting

SSSIA-126	Agentic AI for Regulation and Reporting	CREDITS: 4 Total Hours: 120
Type of Course	Practical	30 hours per credit

Course Objectives

The student will understand and be able to:

- (a) build retrieval-grounded agents over regulatory and accounting source material;
- (b) orchestrate IFRS 17 and solvency calculations through agents with a complete audit trail;
- (c) assess the limits of agent use where regulatory interpretation and professional judgement are required.

Course Syllabus

At the end of the course, the student will be able to:

Unit	Description	Hours
I	1. Retrieval-augmented agents grounded in the NAIC Annual Statement Instructions, Accounting Practices and Procedures Manual (SSAPs) and the relevant ASOPs (20, 36, 41, 43); chunking, indexing and citation so every answer traces to a source paragraph. 2. Why an ungrounded agent must not be used for regulatory or financial reporting questions, demonstrated by example.	22
II	3. Agent-assisted construction and reconciliation of NAIC Annual Statement exhibits, including the Underwriting and Investment Exhibit, Schedule P and Schedule F, with automated tie-outs to source data. 4. Building a regulatory change monitor that flags amendments to statutory accounting guidance affecting a stated practice area.	26
III	5. Risk-Based Capital and IRIS ratio calculation and monitoring agents, producing a solvency dashboard and flagging threshold breaches with the agent explaining each ratio. 6. GAAP, SAP and IFRS difference reconciliation agents, and automated onerous contract testing under IFRS 17 with generation of the supporting evidence.	26
IV	7. Reinsurance accounting agents: risk-transfer testing, Schedule F automation and commutation analysis, with every assertion traceable to a number or a citation. 8. Drafting disclosure and Statement of Actuarial Opinion (SAO) and Actuarial Opinion Summary (AOS) narrative from agent-assembled evidence.	26
V	9. Limits of automation: matters of interpretation, professional judgement and the actuary's personal responsibility for an opinion. 10. Governance of AI use in a regulated reporting process, including model risk, data residency and disclosure to the supervisor.	20
	Total	120

References

- International Accounting Standards Board. IFRS 17 Insurance Contracts.
- International Association of Insurance Supervisors. Insurance Core Principles (current edition).
- International Actuarial Association (2016). IAA Risk Book.
- Model Context Protocol specification and reference servers.

SSSIA-127 Excel for Actuarial Practice - Part I

SSSIA-127	Excel for Actuarial Practice - Part I	CREDITS: 3 Total Hours: 90
Type of Course	Practical	30 hours per credit

Course Objectives

The student will understand and be able to:

- (a) apply the critical steps to model data, document the work, create an audit trail and communicate the approach, results and conclusions effectively to colleagues.

Course Syllabus

At the end of the course, the student will be able to:

Unit	Description	Hours
I	1. Use appropriate tools for checking, cleaning, restructuring and transforming data to make it suitable for analysis. 2. Summarise data using appropriate analysis, descriptive statistics and graphical representation. 3. Select and carry out appropriate statistical tests of reasonableness.	24
II	4. Make appropriate assumptions about the data provided. 5. Repair corrupt or missing data.	24
III	6. Plan and produce a spreadsheet model to solve a specified problem. 7. Create appropriate charts to support visual interpretation of the results.	22
IV	8. Perform checks on the intermediate and final results of a model. 9. Create a clear audit trail that could be followed by a senior actuary and would enable the model to be worked on and corrected by a fellow student, covering key assumptions, description of data and model checks, methodology and reasonableness checks.	20
	Total	90

References

- Walkenbach, J. (2007). Excel Formulas. Wiley. ISBN 978-0470044025.
- Day, A. (2010). Mastering Financial Mathematics in Microsoft Excel, 2nd ed. FT Prentice Hall. ISBN 978-0273730330.
- O'Beirne, P. (2005). Spreadsheet Check and Control: 47 Key Practices to Detect and Prevent Errors. Systems Publishing. ISBN 978-1905404001.
- Polya, G. (1990). How to Solve It. Penguin. ISBN 978-0691150956.

SSSIA-128 Excel for Actuarial Practice - Part II

SSSIA-128	Excel for Actuarial Practice - Part II	CREDITS: 3 Total Hours: 90
Type of Course	Practical	30 hours per credit

Course Objectives

The student will understand and be able to:

- (a) apply the critical steps to model data, maintain the documentation and audit trail, analyse the methods used and the outputs generated, and communicate the approach, results and conclusions effectively to colleagues.

Course Syllabus

At the end of the course, the student will be able to:

Unit	Description	Hours
I	1. Perform checks on the intermediate and final results of a model. 2. Comment on the reasonableness of the results from a model under different scenarios.	24
II	3. Draft a clear summary of the model and the results for a senior actuary, covering the data, assumptions, approach taken, results, conclusions and suggested next steps to develop the model.	24
III	4. Provide commentary on the results from a model appropriate for the target audience, including analytical comments on each stage of the results, explanation of patterns and unusual features, and explanation of the differences between results under the various strategies modelled.	22
IV	5. Identify possible next steps for the client, taking into consideration the initial modelling and the results, including possible enhancements to the model and additional modelling to support the project's objectives.	20
	Total	90

References

- Walkenbach, J. (2007). Excel Formulas. Wiley. ISBN 978-0470044025.
- Day, A. (2010). Mastering Financial Mathematics in Microsoft Excel, 2nd ed. FT Prentice Hall. ISBN 978-0273730330.
- O'Beirne, P. (2005). Spreadsheet Check and Control. Systems Publishing. ISBN 978-1905404001.

SSSIA-129 R for Actuarial Work - Part I

SSSIA-129	R for Actuarial Work - Part I	CREDITS: 3 Total Hours: 90
Type of Course	Practical	30 hours per credit

Course Objectives

The student will understand and be able to:

- (a) produce simple visualisations and statistics from a data set;
- (b) work with the basic properties and uses of commonly used probability distributions and the statistical properties of data generated by random sampling;
- (c) use statistics to make inferences about the process underlying a data set and about relationships between data sets;
- (d) use Bayesian statistics to update prior beliefs about a data-generating process.

Course Syllabus

At the end of the course, the student will be able to:

Unit	Description	Hours
I	1. Describe the purpose and function of data analysis and complete an exploratory data analysis in R. 2. Understand the characteristics of basic univariate distributions and how to generate samples from them.	22
II	3. Determine the characteristics of jointly distributed random variables. 4. Evaluate expectations and conditional expectations, and evaluate and apply generating functions. 5. State and apply the central limit theorem.	24
III	6. Describe random sampling and the sampling distributions of statistics commonly used in statistical inference. 7. Construct estimators and discuss their properties. 8. Calculate confidence intervals and prediction intervals.	22
IV	9. Apply the concepts of hypothesis testing and goodness of fit. 10. Understand and use linear regression models and generalised linear models in R. 11. Explain the fundamental concepts of Bayesian statistics and use these concepts to calculate Bayesian estimators.	22
	Total	90

References

- Peng, R. (2015). Report Writing for Data Science in R. Lean Publishing. ISBN 978-132973364.
- Frees, E. W. (2010). Regression Modelling with Actuarial and Financial Applications. Cambridge University Press. ISBN 978-0521760119.

SSSIA-130 R for Actuarial Work - Part II

SSSIA-130	R for Actuarial Work - Part II	CREDITS: 3 Total Hours: 90
Type of Course	Practical	30 hours per credit

Course Objectives

The student will understand and be able to:

- (a) apply statistical distributions suitable for modelling the variables and risks that arise within insurance contracts;
- (b) apply statistical concepts for modelling, fitting and forecasting data indexed by time;
- (c) apply Markov models to time-indexed risk and claim data;
- (d) describe, estimate and use statistical models for the time until an event occurs.

Course Syllabus

At the end of the course, the student will be able to:

Unit	Description	Hours
I	1. Loss distributions, with and without risk sharing. 2. Compound distributions and their applications in risk modelling. 3. Introduction to copulas and to extreme value theory.	24
II	4. Understand the core concepts underlying time series models. 5. Applications of time series models to insurance data.	22
III	6. Stochastic processes; understand and apply a Markov chain; define and apply a Markov process.	22
IV	7. Concepts of survival models and the estimation procedures for lifetime distributions. 8. Derive maximum likelihood estimators for transition intensities and handle transition intensities dependent on age, by exact or census methods. 9. Graduation and graduation tests; mortality projection.	22
	Total	90

References

- Beyersmann, J., Schumacher, M. and Allignol, A. (2012). Competing Risks and Multistate Models with R. Springer. ISBN 978-1461420347.
- Macdonald, A. S. (1996). An Actuarial Survey of Statistical Models for Decrement and Transition Data. British Actuarial Journal, 2.
- Macdonald, A. S., Richards, S. J. and Currie, I. D. (2018). Modelling Mortality with Actuarial Applications. Cambridge University Press. ISBN 978-1107045415.

SSSIA-131 Python for Actuarial Work - Part I

SSSIA-131	Python for Actuarial Work - Part I	CREDITS: 3 Total Hours: 90
Type of Course	Practical	30 hours per credit

Course Objectives

The student will understand and be able to:

- (a) understand the basics of Python and the usage of the major packages for actuarial data work.

Course Syllabus

At the end of the course, the student will be able to:

Unit	Description	Hours
I	1. Set up the Python environment; basic variable types and how to use them; Python syntax. 2. Introduction to and usage of NumPy.	22
II	3. Introduction to and usage of pandas. 4. Data visualisation with seaborn and matplotlib.	24
III	5. Types of data; handling missing data. 6. Data encoding and feature scaling; handling data imbalance.	22
IV	7. Simple linear regression and multiple linear regression. 8. Polynomial regression.	22
	Total	90

References

- Grus, J. (2019). Data Science from Scratch: First Principles with Python. O'Reilly Media.
- Geron, A. (2019). Hands-On Machine Learning with Scikit-Learn, Keras and TensorFlow. O'Reilly Media.
- Raschka, S. and Mirjalili, V. (2017). Python Machine Learning. Packt Publishing.

SSSIA-132 Python for Actuarial Work - Part II

SSSIA-132	Python for Actuarial Work - Part II	CREDITS: 3 Total Hours: 90
Type of Course	Practical	30 hours per credit

Course Objectives

The student will understand and be able to:

- (a) use Python and its packages to implement machine learning models on insurance data.

Course Syllabus

At the end of the course, the student will be able to:

Unit	Description	Hours
I	1. Decision tree regression and random forest regression. 2. Evaluation of regression models.	20
II	3. K-nearest neighbours; decision tree classification; random forest classification.	20
III	4. Gradient boosting classification. 5. Evaluation of classification models.	20
IV	6. Clustering: K-means, hierarchical and DBSCAN. 7. Principal component analysis.	16
V	8. Basic artificial neural networks. 9. Basic convolutional and recurrent neural networks.	14
	Total	90

References

- Grus, J. (2019). Data Science from Scratch: First Principles with Python. O'Reilly Media.
- Geron, A. (2019). Hands-On Machine Learning with Scikit-Learn, Keras and TensorFlow. O'Reilly Media.
- Raschka, S. and Mirjalili, V. (2017). Python Machine Learning. Packt Publishing.

